

UI and displayers

`EasyIdleGame.UI` supplies optional presentation components for uGUI and helper presenters for UI Toolkit. They are examples, not a required architecture: any UI can bind directly to the public manager and holder APIs described in the [scripting reference](#) (`05-scripting-reference.pdf`). Per-field options are documented in each component's Inspector.

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Supplied uGUI displayers

Assign the definition asset in the Inspector; the holder is resolved at runtime. Displayers require TextMeshPro essentials.

Area	Components
Currencies	CurrencyDisplayer
Businesses and production	BusinessDisplayer, UpgradableBusinessDisplayer, UpgradableBusinessUpgradeDisplayer, CurrentProductionDisplayer
Managers	ManagerDisplayer
Boosts	BoostDisplayer, ActiveBoostDisplayer
Upgrades	UpgradeDisplayer, UpgradesSummaryDisplayer
Buy amount selection	BuyAmountButtonDisplayer
Progression	PlayerLevelDisplayer, PlayerStatsDisplayer, AchievementDisplayer
Daily rewards	DailyRewardListDisplayer, RewardListDayDisplayer
Prestige	PrestigeDisplayer
Openable panels	GenericOpenableDisplayer, BusinessDetailDisplayer, LevelUpDisplayer, ProfitDisplayer — open/close through the <code>IOpenable</code> contract

`BaseListDisplayer` , `BusinessListDisplayer` , `CurrencyListDisplayer` , and `BoostListDisplayer` are obsolete; use Unity layout components instead.

The BaseDisplayer redraw contract

Every displayer derives from `BaseDisplayer` , which separates:

- `Redraw_Default_Empty()` — placeholder-only editor output.
- `Redraw_Default()` — editor output from assigned references.
- `Redraw()` — runtime state.

`autoRedraw` (default on) redraws every frame. Disable it and redraw from manager events in large lists. **Tools > EasyIdleGame > RedrawAll** redraws loaded displayers in the editor.

Subclass a displayer

```

public sealed class MyBusinessDisplayer : BusinessDisplayer
{
    public override void Redraw()
    {
        base.Redraw();
        // Project-specific presentation.
    }
}

```

Call the base `Redraw`, `Redraw_Default`, or `Redraw_Default_Empty` when overriding. Keep economy mutation in managers and holders; UI code presents state and forwards intent.

UI Toolkit presenters

`UIToolkitPresenters.cs` provides `UIToolkitListRowPresenter` plus static presenters for businesses, upgrades, boosts, shop items, and prestige. This is helper code used by the focused feature demos, not a production UI framework — bind UI Toolkit directly to the manager APIs when the helpers do not fit.

Refresh from events

Query UI elements defensively, subscribe after managers and holders exist, and remove listeners on disable or destroy. Refresh labels from amount, unlock, purchase, claim, active-state, and location events; poll only truly time-varying values such as progress bars and countdowns. Use `OnHolderAdded` when definitions can gain holders after your screen is enabled. The event table is in the [scripting reference](#) (`05-scripting-reference.pdf`).

UI helpers and contracts

- `UIUpdater` — static text/image/color update helpers for TMP and uGUI.
- `IOpenable` — open/close contract for panel displayers.
- `IMetadataDisplayer` — binds `Metadata` display name/icon safely.
- `AudioButton` — standard button click feedback; audio setup is in [Configure audio and editor helpers](#) (`03-feature-guides.pdf`).
- `DemoManager` and `SimpleDemoManager` — demo-scene controllers; reference material, not production components.

The same manager APIs and events support full UI Toolkit or uGUI screens at production scale, not just the demo helpers above.